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Database Foundations

6-2

Structured Query Language (SQL)

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Objectives

- This lesson covers the following objectives:
 - Describe how data is organized in a relational database
 - Explain the various relational database terminologies
 - Define the structured query language and its functions
 - Describe how SQL processing takes place
 - Identify the tools used to access the relational database



How Is Data Organized in Relational Databases?

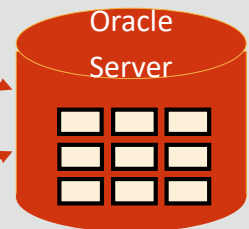
- Data is stored in a two-dimensional matrix known as a table

Table name: EMPLOYEES

DEPARTMENT_ID	DEPARTMENT_NAME	MANAGER_ID
10	Administration	200
20	Marketing	201

Table name: DEPARTMENTS

EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL
100	Steven	King	SKING
101	Neena	Kochhar	NKOCHHAR
102	Lex	De Haan	LDEHAAN



How Is Data Organized in Relational Databases?

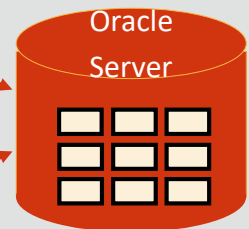
- DBMS software is used to manage reading and manipulating data

Table name: EMPLOYEES

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Oracle software runs on many hardware architectures and operating systems. The computer on which the Oracle Database software resides is known as the Oracle Database server. Additionally, Oracle Database server can refer to the Oracle Database software and its data.

Relational Database Terminology

2	3			4	
EMPLOYEE_ID	FIRST_NAME	LAST_NAME	SALARY	COMMISSION_PCT	DEPARTMENT_ID
100	Steven	King	24000	-	90
101	Neena	Kochhar	17000	-	90
102	Lex	De Haan	17000	-	90
200	Jennifer	Whalen	4400	-	10
205	Shelley	Higgins	12000	-	110
206	William	Gietz	8300	-	110
149	Eleni	Zlotkey	10500	.2	80
174	Ellen	Abel	11000	.3	80
201	Michael	Hartstein	13000	-	20

Note : Refer to notes for explanations to number notations

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A relational database can contain one or many tables. A table is the basic storage structure of an RDBMS. A table holds all data necessary about something in the real world, such as employees, invoices, or customers. The slide shows the contents of the EMPLOYEES table or relation. The numbers indicate the following:

1. A single row (or tuple) representing all the data required for a particular employee. Each row in a table should be identified by a primary key, which permits no duplicate rows. Row order is insignificant; row order can be specified when the data is retrieved.
2. A column or attribute containing the primary key value. The employee_id number identifies a unique employee in the EMPLOYEES table. In this example, the employee_id number column is designated as the primary key. A primary key must contain a value, and the value must be unique.
3. A column that is not a key value. A column represents one kind of data in a table; in this example, the data is the salaries of all the employees. Column order is insignificant when storing data; column order can be specified when the data is retrieved.
4. A column or attribute containing the foreign key value, which is also a foreign key. A foreign key is a column that defines how tables relate to each other. A foreign key refers to a primary key or a unique key in the same table or in another table. In the example, DEPARTMENT_ID uniquely identifies a department in the DEPARTMENTS table.
5. A field can be found at the intersection of a row and a column. There can be only one value in it.
6. A field may have no value in it. This is called a null value. In the EMPLOYEES table, only those employees who

have the role of sales representative have a value in the COMMISSION_PCT (commission) field.

Relating Multiple Tables

- Each row of data in a table can be uniquely identified by a primary key
- You can logically relate data from multiple tables using foreign keys

Table name: **EMPLOYEES**

EMPLOYEE_ ID	FIRST_ NAME	LAST_ NAME	DEPARTMENT_ ID
100	Steven	King	90
101	Neena	Kochhar	90
103	David	Austin	60

↑
Primary key

↑
Foreign key

Table name: **DEPARTMENTS**

DEPARTMENT_ ID	DEPARTMENT_ NAME	MANAGER_ ID
60	IT	103
90	Executive	100

↑
Primary key

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Because data about different entities is stored in different tables, you may need to combine two or more tables to answer a particular question. For example, you may want to know the location of the department where an employee works. In this scenario, you need information from the EMPLOYEES table (which contains data about employees) and the DEPARTMENTS table (which contains information about departments). With an RDBMS, you can relate the data in one table to the data in another by using the foreign keys. A foreign key is a column (or a set of columns) that refers to a primary key in the same table or in another table.

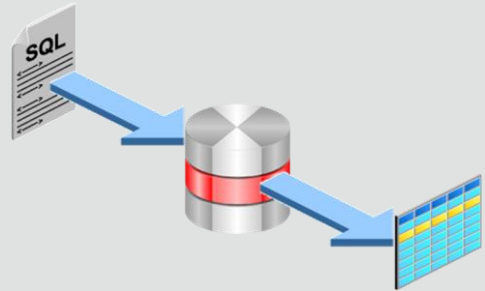
You can use the ability to relate data in one table to data in another to organize information in separate, manageable units. Employee data can be kept logically distinct from the department data by storing it in a separate table.

What Is SQL?

- In a relational database, you do not specify the access route to the tables, and you do not need to know how the data is arranged physically
- To access the database, you execute a SQL statement, which is the American National Standards Institute (ANSI) standard language for operating relational databases
- SQL is also compliant with ISO Standard (SQL 1999)

What Is SQL?

- Structured query language (SQL) is the set-based, declarative language used to access data in an Oracle database
- SQL provides an interface to a relational database and provides statements that help work with the database
- SQL is:
 - Efficient, easy to learn, and use
 - Functionally complete
(With SQL, you can define, retrieve, and manipulate data in the tables)



Functions of SQL

- Creating, replacing, altering, and dropping database objects
- Inserting, updating, and deleting rows in a table
- Querying data stored in the database
- Controlling access to the database and database objects
- Guaranteeing database consistency and integrity



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SQL provide benefits to all types of users:

- Application programmers
- Database administrators
- Managers
- End users

SQL processes sets of data as groups rather than as individual units. It allows users to:

- Access and describe the data stored in the database
- Define the data stored in the database and manipulate that data

Types of SQL Commands

- **DDL** (Data Definition Language) – defines database structures
- **DML** (Data Manipulation Language) – manipulates data (INSERT, UPDATE, DELETE)
- **DQL** (Data Query Language) – SELECTs data
- **DCL** (Data Control Language) – controls user access
- **TCL** (Transactional Control Language) – manages database transactions

SQL Processing

- Stages of SQL processing

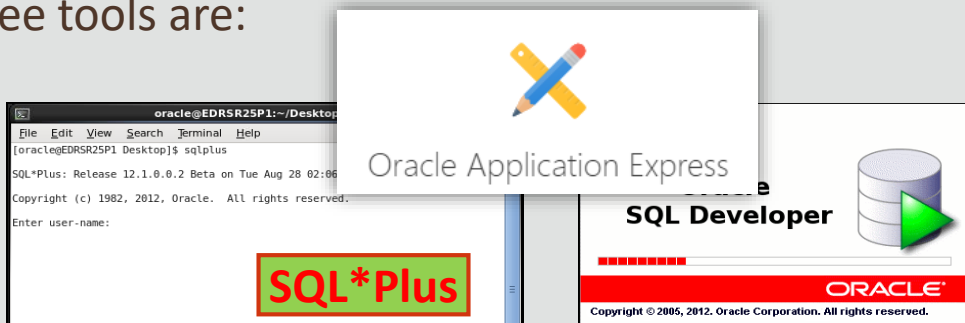


* See definitions of stages in notes

- **SQL Parsing:** The first stage of SQL processing is parsing, which involves separating the pieces of a SQL statement into a data structure that can be processed by other routines. The database parses a statement when it is instructed by the application, which means that only the application, not the database itself, can reduce the number of parses. During the parse, the database performs a syntax check, semantic check, and shared pool check.
- **SQL Optimization:** It is the process of selecting the most efficient means of executing a SQL statement. The database optimizes queries based on statistics collected about the actual data being accessed. The optimizer uses the number of rows, the size of the data set, and other factors to generate possible execution plans, assigning a numeric cost to each plan. The database uses the plan with the lowest cost.
- **SQL Row Source Generation:** The row source generator is software that receives the optimal execution plan from the optimizer and produces an iterative plan, called the query plan, that is usable by the rest of the database. The iterative plan is a binary program that, when executed, produces the result set. The query plan takes the form of a combination of steps. Each step returns a row set. The rows in this set are either used by the next step or, in the last step, are returned to the application issuing the SQL statement.
- **SQL Execution:** The SQL engine executes each row source in the tree produced by the row source generator.

Accessing Data in the Oracle Database Server

- Specialized tools installed on users' computers allow them to access data on the Oracle Database server. These tools are called clients and are used to send SQL instructions (commands) to the server
- Three tools are:



* Note: In this class we will be using Oracle Application Express

Case Scenario: Need to Extract Data



Faculty

Sean, by now you should understand the need to use SQL to access data in a relational database. Can you think of scenarios where retrieving data from a database table would be necessary?

Sure. Let me think of some use cases



Student

Use Cases

- Think of and discuss the use of databases to store data and produce information in these and other areas



Hospitals



Retail



Airports



Schools

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The need to store and manipulate data arises in different sectors of industry. Examples include Health Services, Transportation, Retail, and Education.

- Patient data is stored in a hospital and is called a medical record.
- Airlines store data for passengers so that ticketing and checkin can be completed online.
- Schools maintain student and teacher records so that course records can be retrieved for transcripts.

Summary

- In this lesson, you should have learned how to:
 - Describe how data is organized in a relational database
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